MEMORANDUM

TO: Dr. Tina Han

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CC: ME32301 Lab Section 08- Harbin, Zachary; Alshaikh, Leen

DATE: January-28-2025

RE: Poisson's Ratio Lab 1

EXECUTIVE SUMMARY

The main purpose of this experiment was to measure the Poisson's ratio for three different materials, namely, steel, brass, and copper. For testing purposes, each of the three specimens was attached to two strain gauges in Wheatstone Bridge configuration. One for measuring the axial strain and another for lateral strain measurement. These were then connected to the Instron Universal Test Machine, where the Data Acquisition System (DAS) is situated. Each metal specimen was then subjected to uniaxial tensile loading, recording axial and lateral strains at defined increments of force. After one trial per specimen, data were collected and subjected to linear regression analysis in order to calculate the Poisson's ratio for each specimen.

The following table provides an overview of the tested specimens and applied loading conditions as presented in Table 1 below.

Test	**Specimen**	D[in]	W[in]	L[in]	**Maximum Force [kN]**	**Recording Increment [kN]**
1	Brass	0.25	0.25	8	3	0.3
2	Steel	0.25	0.25	8	3	0.3
3	Copper	0.25	0.25	8	3	0.3

Table 1: Test Specifications

Possion's ratio was calculated by analyzing the gradient of the lateral strain versus axial strain data using Equation (1). The intercepts of the linear regression models were close to zero, as expected, since the absence of axial strain must be accompanied by the absence of lateral strain. Furthermore,

a 95% confidence interval was calculated to determine the reliability of the experimental results.

$$\nu = -\frac{\varepsilon_L}{\varepsilon_A} \tag{1}$$

Table 2 provides the experimental values for the Poisson's ratio along with the published literature values for each tested specimen, where ν_{exp} stands for the experimental Poisson's ratio and ν_{pub} stands for the published Poisson's ratio value.

Specimen	ν_{exp}	ν_{pub}	**Percent Error**
Brass	0.266 ± 0.00007	0.331	19.6
Steel	0.274 ± 0.00027	0.265	3.4
Copper	0.284 ± 0.00014	0.355	20

Table 2: Experimental and Published Poisson's Ratios with Percent Error

The results of the experiment indicated sources of random as well as systematic error. A few factors that could have affected the accuracy of the data include some degree of misalignment and imperfect setup of the strain gauge, random vibrations from surroundings, and the movement of experimental wires associated with the device. Furthermore, the specimens went through multiple loading cycles which could have reduced the bond strengths at the molecular level, causing slight weakening and thus impacting results' accuracy. To minimize such inconsistencies, it is important to use specimens that are in pristine condition, prior to testing.

Reflecting on the experiment, a major aspect was to ensure precise calibration of the strain gauges and load cells. A few suggestions include, repeating tests and averaging results could help minimise individual anomalies. Secondly, limiting the testing area to a single person could help minimize vibrations and ensure greater consistency in procedures. However, despite the deviations from the literature values, the experiment provided deep insights into the factors that affect experimental accuracy, providing possible methods of having more accurate strain data in order to produce a more precise Poisson's ratio.

RESULTS

Three metal specimens were tested to obtain their Poisson's ratios. The experimental Poisson's ratio was determined by finding the slope of the linear model after graphing the axial and lateral strain of the specimens, as shown in the graphs below:

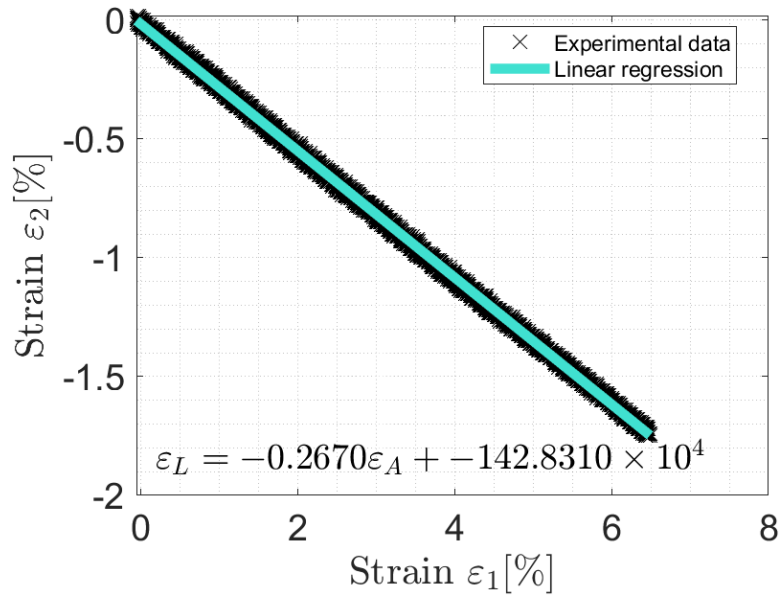

Figure 1: Lateral Stress vs. Axial Stress for Brass sample

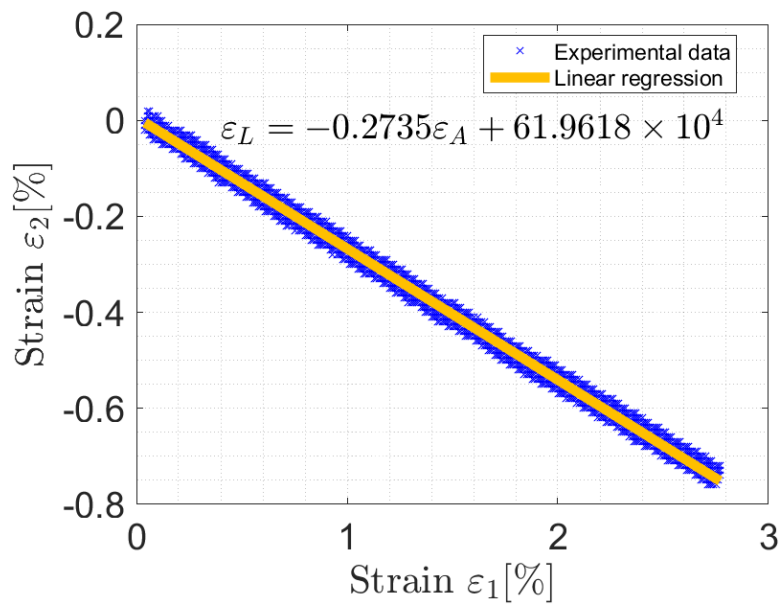

Figure 2: Lateral Stress vs. Axial Stress for Steel sample

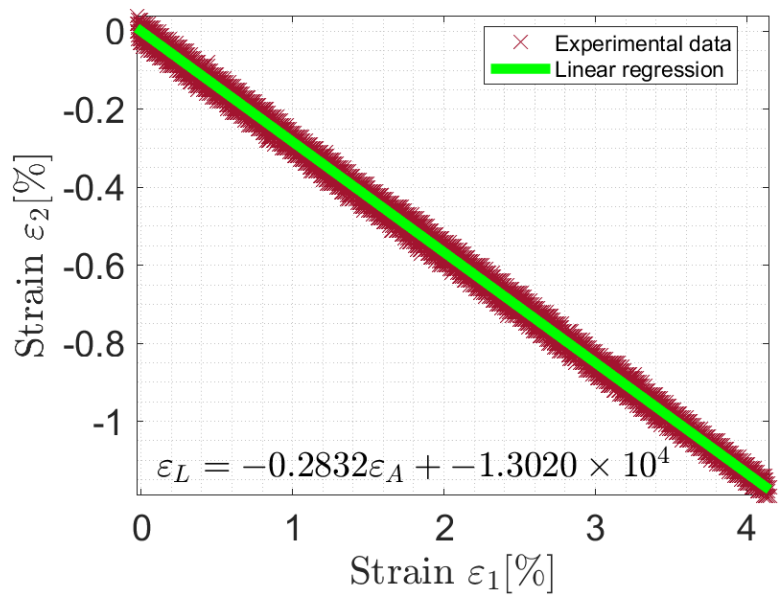

Figure 3: Lateral Stress vs. Axial Stress for Copper sample

The intercepts of the linear regression models represent lateral strain when there is no axial strain. They are close to zero, which makes sense because there should not be any lateral strain without axial strain.

Linear regression was also performed on the results to obtain the 95% confidence interval, which forms the margin of error of the experimental results:

a.

	Coefficients	andard Erro	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	0.006109	0.000161	38.04686	0	0.005794	0.006423	0.005794	0.006423
Slope	-0.28385	6.97E-05	-4074.06	0	-0.28399	-0.28372	-0.28399	-0.28372

b.

	Coefficients	andard Erro	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	0.007646	0.000219	34.97963	4.6E-261	0.007217	0.008074	0.007217	0.008074
Slope	-0.27395	0.000137	-1999.22	0	-0.27422	-0.27368	-0.27422	-0.27368

c.

	Coefficients	andard Erro	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	-0.01944	0.000134	-145.018	0	-0.0197	-0.01918	-0.0197	-0.01918
Slope	-0.26648	3.62E-05	-7370.44	0	-0.26655	-0.26641	-0.26655	-0.26641

Figure 4: Linear regression results of the three specimens: a. copper; b. steel; c. brass

The results of the tests are detailed in Table 3. ν_{exp} represents the experimental value obtained through testing, while ν_{pub} represents the established value [1].

Specimen	ν_{exp}	ν_{pub}
Brass	0.266 ± 0.00007	0.331
Steel	0.274 ± 0.00027	0.265
Copper	0.284 ± 0.00014	0.355

Table 3: Experimental and published values of Poisson's ratios for specimens.

The experimental Poisson's ratio of all three specimens differs from the published values. As can be seen in Table 4, the experimental Poisson's ratio of brass is 19.6% lower than the published value. The experimental value for steel is 3.4% higher than the published one. The experimental Poisson's ratio of copper is 20% higher than the published value. The percent error formula used to calculate these values can be seen in Equation 2.

$$PercentError = \frac{|V_{\text{actual}} - V_{\text{measured}}|}{V_{\text{actual}}} \tag{2}$$

Specimen	Percent Error
Brass	19.6
Steel	3.4
Copper	20

Table 4: Percent Error of Poisson's ratios of specimens compared published values.

CONCLUSIONS

The experiments were conducted to establish the Poisson's ratio for three types of metal specimens- steel, brass, and copper. Data analyses were made through linear regression following the principle stated in Equation (1), where Poisson's ratio is the negative lateral strain divided by axial strain. The experiment was conducted on each of these three materials and their results are plotted along with some published values in Table 3. The best fit from the data using a linear regression analysis yielded a 95% confidence interval of the measured value with 0.266 ± 0.00007 for brass, 0.274 ± 0.00027 for steel, 0.284 ± 0.00014 for copper. The highest percent error obtained was that for steel, which was 3.4%. The second lowest was obtained for brass at 19.6%, while copper was 20% as obtained in Table 4.

High percent errors indicated that the Poisson's ratio based on data collected from this experiment did not agree with the published values. These discrepancies could have resulted from a number of factors. The metal samples tested were recycled ones that could easily result in their failing to correspond with the position of the stress and strain gauges. Furthermore, the lab's temperature was not regulated to eliminate material contraction and expansion due to which the materials could have undergone intermolecular changes. Besides, within the lab, there were also more than 20 people conducting experiments, making noise, or working simultaneously where many machines faced running conditions; this could potentially vibrate testing equipment. Even slight movements in machine wires can be caused by people moving and can affect such sensitive components in their results.

In light of these errors, further experiments should be conducted in a controlled environment with a single person in the vicinity of the equipment. Materials must be precisely calibrated before testing for proper alignment. More tests using different samples of the same materials should be done to increase the accuracy.

REFERENCE LIST

[1] E. Toolbox. "Poisson's ratio." (2008), [Online]. Available: https://www.engineeringtoolbox.com/poissons-ratio-d_1224.html (visited on 01/29/2025).

[2] OpenAI. "ChatGPT (February 7, 2025 Version)," [Online]. Available: https://openai.com/chat-gpt (visited on 02/07/2025).

DATA SHEETS

These screenshots display the raw data utilized in this experiment for the various metal specimens.

2025_SP_Section08_Team04_Lab01_Steel

Time	Displacement	Force	STRAIN2	STRAIN1
(s)	(mm)	(kN)	(%)	(%)
0.0000	0.0000	0.0081	-0.0003	0.0007
0.0200	0.0000	0.0082	-0.0003	0.0007
0.0400	0.0000	0.0082	-0.0003	0.0006
0.0600	0.0000	0.0082	-0.0003	0.0006
0.0800	0.0000	0.0082	-0.0003	0.0006
0.1000	0.0000	0.0079	-0.0003	0.0007
0.1200	0.0000	0.0079	-0.0003	0.0007
0.1400	0.0000	0.0081	-0.0003	0.0006
0.1600	0.0001	0.0081	-0.0003	0.0006
0.1800	0.0001	0.0084	-0.0003	0.0007
0.2000	0.0000	0.0086	-0.0003	0.0007
0.2200	0.0001	0.0084	-0.0003	0.0007
0.2400	0.0001	0.0083	-0.0003	0.0006
0.2600	0.0001	0.0083	-0.0003	0.0006
0.2800	0.0001	0.0082	-0.0003	0.0007
0.3000	0.0001	0.0083	-0.0003	0.0007

Steel Data Sheet

2025_Sp_Section08_Team4_Lab01_Brass_3[67

Time	Displacement	Force	STRAIN2	STRAIN1
(s)	(mm)	(kN)	(%)	(%)
0.0000	0.0000	0.0012	-0.0001	-0.0002
0.0200	0.0000	0.0013	-0.0001	-0.0002
0.0400	0.0000	0.0013	0.0000	-0.0003
0.0600	0.0000	0.0013	0.0001	-0.0004
0.0800	0.0000	0.0011	0.0000	-0.0003
0.1000	0.0000	0.0009	0.0000	-0.0003
0.1200	0.0000	0.0009	0.0000	-0.0004
0.1400	0.0000	0.0008	0.0001	-0.0005
0.1600	0.0000	0.0009	-0.0001	-0.0003
0.1800	0.0000	0.0012	-0.0001	-0.0003
0.2000	0.0000	0.0013	0.0000	-0.0004
0.2200	0.0000	0.0013	-0.0001	-0.0004
0.2400	0.0000	0.0013	-0.0001	-0.0004
0.2600	0.0001	0.0011	-0.0001	-0.0003
0.2800	0.0001	0.0009	-0.0001	-0.0004
0.3000	0.0001	0.0010	0.0000	-0.0005

Brass Data Sheet

2025_Sp_Section08_Team4_Lab01_Copper_28

Time	Displacement	Force	STRAIN2	STRAIN1
(s)	(mm)	(kN)	(%)	(%)
0.0000	0.0000	0.0007	0.0000	-0.0002
0.0200	0.0000	0.0006	0.0000	-0.0003
0.0400	0.0000	0.0006	0.0000	-0.0003
0.0600	0.0000	0.0006	0.0000	-0.0002
0.0800	0.0000	0.0008	0.0000	-0.0002
0.1000	0.0000	0.0010	-0.0001	-0.0001
0.1200	0.0000	0.0010	-0.0001	-0.0002
0.1400	0.0000	0.0010	-0.0001	-0.0002
0.1600	0.0000	0.0009	0.0000	-0.0002
0.1800	0.0000	0.0008	-0.0001	-0.0001
0.2000	0.0001	0.0008	-0.0001	-0.0001
0.2200	0.0001	0.0008	0.0000	-0.0001
0.2400	0.0001	0.0009	0.0001	-0.0002
0.2600	0.0001	0.0010	0.0002	-0.0003
0.2800	0.0001	0.0012	0.0002	-0.0003
0.3000	0.0001	0.0012	0.0000	-0.0001

Copper Data Sheet

APPENDIX

The MATLAB script Brass.m is designed to analyze strain data from an experimental dataset and apply linear regression modeling to examine the relationship between axial and lateral strain in a brass specimen.

```
%%%
%You will need Statistics and Machine Learning Toolbox installed on Matlab
%to run this file. If you do not have the Toolbox installed, type "fitlm"
%in the Command Window, Matlab will prompt you to install the Toolbox
%necessary.
%%%
clc          % Clean up Command Window
clear all    % Clean up all stored variables
close all    % Close all open windows

%%%
% Do not change, this list includes 9 preset colors
% Feel free to use these colors in your plot
colorlist=["#6c3483","#CCCCFF", "#6495ED","#40E0D0","#9FE2BF",...
    "#DE3163","#FF7F50","#FFBF00","#f7dc6f"];

%%%
%Edit this portion, Use readmatrix function to import axial and laterial
%strain data
data=readmatrix("2025_Sp_Section08_Team4_Lab01_Brass_3.csv");

data_ax_strain=data(:,5)*100; % Axial Strain Data in percentage
data_la_strain=data(:,4)*100; % Lateral Strain Data in percentage

%%%
%Do not change, mdl is the linear regression model generated automatically
```

```
%by Matlab. You can find documentation here: https://www.mathworks.com/help/stats/fitlm.html
mdl=fitlm(data_ax_strain,data_la_strain,'VarNames',{'Axial_Strain','Lateral_Strain'});

%%%
%Do not change, this section is used to generate the plot with both
%experimental data and linear regression model.
figure(1)
hold on
mdl_plot=plot(mdl);
delete(mdl_plot(3));

%%%
%Change this section as necessary. mdl_plot(1) is the experimmental data
%and mdl_plot(2) is the linear regression fit. When choosing marker type
%and colors make sure they are easy to read when inserted in the Memo. You
%can find documentation here: https://www.mathworks.com/help/matlab/ref/plot.html
mdl_plot(1).MarkerSize=10;
mdl_plot(1).Marker="x";
mdl_plot(1).Color="k";
mdl_plot(1).MarkerFaceColor="auto";
mdl_plot(2).LineStyle="-";
mdl_plot(2).LineWidth=5;
mdl_plot(2).Color=colorlist(4);

%%%
%This section of the code adds the linear fit equation to the plot. Note that the
%intersection point is denoted by scientific notation due to its small
%magnitude. If you want to change the magnitude of scientific notation, you
%need to edit both within sprintf(). In case you haven't noticed it, the
```

```
%equation is typed with Latex notation too (except using \\ instead of \).
%Depends on the data, you will need to move the location of the annotation.
%To do so, edit the first two values in the text() function.
Eq_text=sprintf('$\\varepsilon_L = %.4f \\varepsilon_A + %.4f\\times 10^{4}$'
    ...
        , mdl.Coefficients.Estimate(2), mdl.Coefficients.Estimate(1)*10000);
Eq_Anno=text(0.2,-1.85,Eq_text,'Interpreter','latex','FontName','Arial','
    FontSize',18);

%%%
%Do not change this section
box on
grid minor
title('');

%%%
%Edit this section to include proper axis label and legend. Current one is
%not acceptable, just purely to demonstrate how to use Latex interpreter if
%necessary.
xlabel('Strain $\varepsilon_{1}$[\%]','interpreter','latex');
ylabel('Strain $\varepsilon_{2}$[\%]','interpreter','latex');

legend('Experimental data','Linear regression','fontsize',11)

%%%
%Feel free to change the font size and font name as needed. Generally I
%recommend Arial for font. Take into consideration that when you insert
%the graph into the figure, the fonts should still be comparable to the
%text font size.
ax = gca;
ax.FontName='Arial';
ax.FontSize=18;

```

```

%%%
%Do not change. This part of the code automatically output the results from
%Linear Regression model in text format. If you are using Matlab for the
%model, take a screeshot of the command window with the output and use it
%in place of the Excel out put on Page Lab Session 1-8.
CI=coefCI(mdl);
disp(mdl);
fprintf('\n95%% Confidence Interval based on the %i data points are:\n',length
    (data_ax_strain));
fprintf('for Intercept: [%.6E, %.6E]\n',CI(1,1),CI(1,2));
fprintf('for Slope:     [%.6f, %.6f]\n',CI(2,1),CI(2,2));

%%%
% Uncomment the following line is you want to automatically save the figure
% as a .png file.
exportgraphics(gcf,'BrassFigure.png');
```